Solution Requirements

Date	23 October 2026
Team ID	[Your Team ID]
Project Name	TravelMate
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users sign up and sign in with email and password through Supabase Auth, with input validation, loading states, error/success messages and Sign Out.	High
2	Authorization levels	A single user role: an authenticated user can create, view, edit and delete only their own trips. Access is enforced with Supabase Row Level Security and backend checks.	High
3	External interfaces	Groq API (AI itinerary generation), Supabase (authentication and database), Leaflet with OpenStreetMap (maps), Nominatim (geocoding) and Google Maps links (directions).	High
4	Transaction processing	Create, read, update and delete trip records; send trip preferences to the backend and receive a structured itinerary (days and activities). No payments are processed.	High
5	Reporting	Itinerary PDF export (A4, multi-page) and a budget summary showing planned budget, activity cost total, remaining budget, daily average and day-wise cost.	Medium
6	Business rules, etc.	A trip needs a destination, valid dates and a budget. Missing or invalid activity costs are handled safely, and the app flags within-budget and over-budget states.	High
7	Compliance to laws or regulations	Protect user data and credentials: secret keys stay on the backend, .env files are not committed, and AI-generated prices and timings are shown as estimates.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
8	Other	Search and sort trips, regenerate an itinerary, show or hide the map, and provide a responsive and accessible interface.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages and dashboard views load quickly; the UI shows loading indicators during AI generation.	Page load under 3 seconds; itinerary generation feedback shown immediately (to be measured)
2	Scalability	The backend is stateless so it can run more instances; Supabase handles hosted data storage.	Supports concurrent users on hosted plans (target to be validated in load testing)
3	Security & Data Privacy	API keys stay on the server, HTTPS is used, CORS is limited to the frontend origin and RLS protects user data.	No private key in the frontend bundle; RLS enabled on trip tables
4	Reliability & Availability	A /health endpoint reports backend status, geocoding failures fall back gracefully, and errors are shown to the user.	Health check returns {"status":"ok"}; no crash on missing coordinates
5	Usability & Accessibility	Responsive design, focus-visible styling, accessible labels on action buttons and mobile navigation.	No horizontal overflow at 375px; targets WCAG 2.1 AA
6	Other	Maintainability through separate components, services, routes and schemas; portability through environment-based configuration.	Frontend and backend configured only via environment variables